



APPENDIX A

HAZMAT ASSESSMENT



HAZARDOUS MATERIALS SUMMARY REPORT

Roof Replacement Project

**Town Of Amherst
Amherst Pelham Regional Middle School
170 Chestnut Street
Amherst, Massachusetts**

July 2025

Prepared for:

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Positive ACM Sample Locations

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1.0 INTRODUCTION

CDW Consultants, Inc. (CDW) conducted a hazardous building materials (HBM) survey of the roofs at the Amherst Pelham Regional Middle School located at 170 Chestnut in Amherst, Massachusetts (Site). The HBM survey was conducted by CDW in support of the proposed roof renovations.

On June 30, July 1 and 2, 2025, Bryant Dana (Massachusetts Asbestos Inspector #AI901229) conducted an inspection for suspect asbestos containing materials (ACM). An inspection is required by the United States Environmental Protection Agency (USEPA) National Emission Standards for Hazardous Air Pollutants (NESHAP), prior to scheduled building renovations or demolitions. Additionally, the Massachusetts Department of Environmental Protection (MassDEP) regulates ACM associated with renovation, demolition, and asbestos abatement projects.

2.0 PROJECT UNDERSTANDING

The scope of the work was to provide an HBM survey in support of the proposed roof renovations at the Amherst Pelham Regional Middle School. The HBM survey consisted of a visual survey of all visually accessible areas, to identify suspect building materials and perform limited sampling, including roof cuts and miscellaneous sampling, for potential hazardous materials samples for analytical analysis for ACM and LBP. The HBM survey included all accessible roofs.

3.0 GENERAL SITE CONDITIONS

The roof at the Amherst Pelham Regional Middle School has approximately 129,000 square feet of roofs. The Amherst Pelham Regional Middle School is separated into three sections. The three sections consist of the recreation building (pool/gyms), auditorium, and the main school building. Throughout the three roof sections, metal decking was primarily used with concrete decking present in a few locations.

4.0 REPORTS REVIEW

There were no previous reports available for review.

5.0 ASBESTOS SURVEY

5.1 Methods

The investigative work for the asbestos survey included conducting a visual inspection of physically accessible areas of the subject areas. Once the visual inspection was completed, the building components were categorized into homogeneous areas. CDW collected limited bulk samples of different homogeneous suspect materials for asbestos analysis. The bulk samples were delivered under chain of custody to Asbestos Identification Laboratory, Inc. (AIL) located in Woburn, Massachusetts. AIL is a state licensed (AA000208) and NVLAP-accredited laboratory

(lab code #200919-0) for asbestos analysis. Bulk samples were analyzed for asbestos content by polarized light microscopy (PLM) using USEPA Method 600/R-93/116. A positive stop method was used – if one sample in a homogeneous group is positive then additional samples of the same material are not analyzed. Samples analyzed to contain greater than 1% asbestos are to be treated as ACM as defined by the USEPA and Commonwealth of Massachusetts Department of Environmental Protection (MassDEP). Photographs are provided in Appendix A. The asbestos analytical report is provided in Appendix B.

5.2 Findings

Findings are listed in the table below.

Sample ID	Material	Location	Asbestos%
1A,B	Dark Gray Caulk on Flashing	Roof over Swimming Pool/Gyms	ND
2A,B	Gray Caulk (Expansion Joint)	Roof over Swimming Pool/Gyms	ND
3A,B	Gray Caulk on Louver	Roof over Swimming Pool/Gyms	ND
4A,B	Dark Gray Caulk on Flashing of Chimney	Roof over Swimming Pool/Gyms	ND
5A,B	Excess Tar on Brick Wall	Roof over Swimming Pool/Gyms	ND
6A,B	White Caulk on Sides (Ends) of Flashing	Roof over Swimming Pool/Gyms	ND
7A,B	White Sealant on HVAC Vent Hood	Roof over Swimming Pool/Gyms	ND
8A,B	Dark Gray Caulk on HVAC Vent Hood	Roof over Swimming Pool/Gyms	ND
9A,B	Gray Sealant on Gutter	Roof over Auditorium	Chrysotile 2%
10A,B,C	Gray Caulking on Flashing	Roof over Auditorium	ND
11A,B	Dark Gray Caulk Under Gray Caulk on Flashing (in some locations)	Roof over Auditorium	ND
12A,B	White Caulk on Flashing	Roof over Auditorium	ND
13A,B	Gray Expansion Joint	Roof over Auditorium	ND
Roof Core 1A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND

Sample ID	Material	Location	Asbestos%
Roof Core 1B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 1C	Tar	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 2A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 2B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 3A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 3B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 3C	Tar	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 4A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 4B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 4C	Tar	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 5A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 5B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 5C	Tar/Fiberboard	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 6A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 6B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 6C	Tar/Fiberboard	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 7A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 7B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 7C	Tar/Fiberboard	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 8A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 8B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 8C	Tar	(Pool/Gyms)	ND

Sample ID	Material	Location	Asbestos%
		Top of Metal Deck	
Curb Cut 1A	Glue/Adhesive on Rubber	(Pool/Gyms) Top Layer	ND
Roof Core 9A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 9B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 10A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 10B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Roof Core 10C	Tar	(Pool/Gyms) Top of Metal Deck	ND
Roof Core 11A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) Top Layer	ND
Roof Core 11B	Tar Paper	(Pool/Gyms) Top/Bottom of Foam	ND
Curb Cut 2A	Glue/Adhesive on Rubber	(Pool/Gyms) (Chimney) Top Layer	ND
Roof Core 12A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 12B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 13A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 13B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 14A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 14B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 15A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 15B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 16A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 16B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 16C	Tar on Concrete Deck	(Auditorium) Top of Concrete Deck	ND

Sample ID	Material	Location	Asbestos%
Roof Core 17A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 17B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 18A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 18B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 18C	Tar on Concrete Deck	(Auditorium) Top of Concrete Deck	ND
Curb Cut 3A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 3B	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Curb Cut 4A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 4B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 4C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 19A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 19B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 19C	Tar/Fiberboard	(Auditorium) Top of Metal Deck	ND
Roof Core 20A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 20B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 20C	Tar/Fiberboard	(Auditorium) Top of Metal Deck	ND
Roof Core 21A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 21B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 21C	Tar/Fiberboard	(Auditorium) Top of Metal Deck	ND
Curb Cut 5A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 5B	Insulation	(Auditorium) Between Rubber and	ND

Sample ID	Material	Location	Asbestos%
		Wall	
Curb Cut 5C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 22A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 22B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 22C	Tar	(Auditorium) Top of Deck	ND
Roof Core 23A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 23B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 24A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 24B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 24C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Curb Cut 6A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 6B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 6C	Tar (Vapor Barrier)	(Auditorium) On Wall	Chrysotile 5%
Curb Cut 7A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 7B	Insulation	(Auditorium) Between Rubber and Wall	ND
Roof Core 25A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 25B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 25C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Roof Core 26A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 26B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 26C	Tar/Fiberboard	(Auditorium) Top of Deck	ND

Sample ID	Material	Location	Asbestos%
Curb Cut 8A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 8B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 8C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 27A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 27B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 27C	Tar/Fiberboard	(Auditorium) Top of Deck	
Roof Core 28A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 28B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 28C	Tar/Fiberboard	(Auditorium) Top of Deck	Chrysotile <1%
Roof Core 29A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 29B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 29C	Tar	(Auditorium) Top of Concrete Deck	ND
Roof Core 30A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 30B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 30C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Roof Core 31A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 31B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 31C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Curb Cut 9A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 9B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 9C	Tar (Vapor Barrier)	(Auditorium)	ND

Sample ID	Material	Location	Asbestos%
		On Wall	
Curb Cut 10A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 10B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 10C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 32A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 32B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 32C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Roof Core 33A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 33B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 33C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Curb Cut 11A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 11B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 11C	Tar (Vapor Barrier)	(Auditorium) On Wall	Chrysotile 20%
Roof Core 34A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 34B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 34C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Curb Cut 12A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 12B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 12C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 35A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND

Sample ID	Material	Location	Asbestos%
Roof Core 35B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 35C	Tar/Fiberboard	(Auditorium) Top of Deck	ND
Roof Core 36A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 36B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 36C	Tar	(Auditorium) Top of Concrete Deck	ND
Roof Core 37A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 37B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 37C	Tar	(Auditorium) Top of Concrete Deck	ND
Curb Cut 13A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 13B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 13C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Curb Cut 14A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 14B	Insulation	(Auditorium) Between Rubber and Wall	ND
Curb Cut 14C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 38A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 38B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 38C	Brick	(Auditorium) Top of Deck	ND
Roof Core 38D	Tar on Brick	(Auditorium) Top of Deck	ND?
Roof Core 38E	Tar/Fiberboard	(Auditorium) Top of Metal Deck	ND
Curb Cut 15A	Glue/Adhesive on Rubber	(Auditorium) Top Layer	ND
Curb Cut 15B	Insulation	(Auditorium)	ND

Sample ID	Material	Location	Asbestos%
		Between Rubber and Wall	
Curb Cut 15C	Tar (Vapor Barrier)	(Auditorium) On Wall	ND
Roof Core 39A	Glue/Adhesive on Rubber Membrane	(Auditorium) Top Layer	ND
Roof Core 39B	Tar Paper	(Auditorium) Top/Bottom of Foam	ND
Roof Core 40A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 40B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Curb Cut 16A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Roof Core 41A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 41B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 41C	Tar	(Main School) Top of Deck	ND
Roof Core 42A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 42B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 42C	Tar	(Main School) Top of Deck	ND
Roof Core 43A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 43B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Curb Cut 17A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Curb Cut 17B	Tar on Wood	(Main School) On Wall	ND
Roof Core 44A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 44B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 44C	Tar	(Main School) Top of Deck	ND
Curb Cut 18A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Roof Core 45A	Glue/Adhesive on	(Main School)	ND

Sample ID	Material	Location	Asbestos%
	Rubber Membrane	Top Layer	
Roof Core 45B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Curb Cut 19A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Curb Cut 20A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Curb Cut 21A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Roof Core 46A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 46B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 47A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 47B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 47C	Tar	(Main School) Top of Metal Deck	ND
Roof Core 48A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 48B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 48C	Tar	(Main School) Top of Deck	Chrysotile 3%
Roof Core 49A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 49B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 50A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 50B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 50C	Tar	(Main School) Top of Deck	Chrysotile 2%
Roof Core 51A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 51B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 51C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 52A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND

Sample ID	Material	Location	Asbestos%
Roof Core 52B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 52C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 53A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 53B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 54A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 54B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 54C	Tar/Fiberboard	(Main School) Top of Deck	ND
Curb Cut 22A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Roof Core 55A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 55B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 55C	Tar/Fiberboard	(Main School) Top of Deck	ND
Curb Cut 23A	Glue/Adhesive on Rubber	(Main School) Top Layer	ND
Curb Cut 23B	Tar (Vapor Barrier)	(Main School) On Wall	ND
Roof Core 56A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 56B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 57A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 57B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Curb Cut 24A	Glue/Adhesive on Rubber	(Main School) Top Layer (Rubber Membrane to Brick)	ND
Curb Cut 25A	Glue/Adhesive on Rubber	(Main School) Top Layer (Wood Wall)	ND
Roof Core 58A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 58B	Tar Paper	(Main School)	ND

Sample ID	Material	Location	Asbestos%
		Top/Bottom of Foam	
Roof Core 58C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 59A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 59B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 59C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 60A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 60B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 60C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 61A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 61B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 61C	Tar/Fiberboard	(Main School) Top of Deck	ND
Curb Cut 26A	Glue/Adhesive on Rubber	(Main School) On Chimney (Rubber to Wood)	ND
Roof Core 62A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 62B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 63A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 63B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Curb Cut 27A	Glue/Adhesive on Rubber	(Main School) Top Layer (Rubber Membrane to Brick)	ND
Curb Cut 28A	Glue/Adhesive on Rubber	(Main School) Top Layer (On Wood)	ND
Roof Core 64A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 64B	Tar Paper	(Main School) Top/Bottom of Foam	ND

Sample ID	Material	Location	Asbestos%
Roof Core 64C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 65A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 65B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 65C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 66A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 66B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 66C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 67A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 67B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 67C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 68A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 68B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 68C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 69A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 69B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 69C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 70A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 70B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 70C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 71A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 71B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 71C	Tar/Fiberboard	(Main School) Top of Deck	ND

Sample ID	Material	Location	Asbestos%
		Deck	
Roof Core 72A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 72B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 72C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 73A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 73B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 73C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 74A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 74B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 74C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 75A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 75B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 76A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 76B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 76C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 77A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 77B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 77C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 78A	Glue/Adhesive on Rubber Membrane	(Main School) Top Layer	ND
Roof Core 78B	Tar Paper	(Main School) Top/Bottom of Foam	ND
Roof Core 78C	Tar/Fiberboard	(Main School) Top of Deck	ND
Roof Core 79A	Glue/Adhesive on Rubber Membrane	(Main School) (School Entrance)	ND

Sample ID	Material	Location	Asbestos%
		Top Layer	
Roof Core 79B	Tar Paper	(Main School) (School Entrance) Top/Bottom of Foam	ND
Roof Core 79C	Tar/Fiberboard	(Main School) (School Entrance) Top of Deck	ND
Roof Core 80A	Glue/Adhesive on Rubber Membrane	(Main School) (School Entrance) Top Layer	ND
Roof Core 80B	Tar Paper	(Main School) (School Entrance) Top/Bottom of Foam	ND
Roof Core 80C	Tar/Fiberboard	(Main School) (School Entrance) Top of Deck	ND
Roof Core 81A	Glue/Adhesive on Rubber Membrane	(Main School) (School Entrance) Top Layer	ND
Roof Core 81B	Tar Paper	(Main School) (School Entrance) Top/Bottom of Foam	ND
Roof Core 81C	Tar/Fiberboard	(Main School) (School Entrance) Top of Deck	ND
Roof Core 82A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) (Over Boiler Room) Top Layer	ND
Roof Core 82B	Tar Paper	(Pool/Gyms) (Over Boiler Room) Top/Bottom of Foam	ND
Roof Core 83A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) (Over Boiler Room) Top Layer	ND
Roof Core 83B	Tar Paper	(Pool/Gyms) (Over Boiler Room) Top/Bottom of Foam	ND
Curb Cut 29A	Glue/Adhesive on Rubber Membrane	(Pool/Gyms) (Over Boiler Room) Top Layer	ND
14A,B,C	Dark Gray Sealant	Under Finishing over Auditorium Entrance	ND
15A,B	Dark Gray Sealant	On Roof Drains	ND

Sample ID	Material	Location	Asbestos%
		(Auditorium)	
16A,B	Tar	On Hatch (Auditorium)	ND
17A,B	Dark Gray Caulk	On Flashing (Main School Roof Chimney)	ND
18A,B	Dark Gray Caulk	On Flashing (Main School Roof Above Library)	ND
19A,B	Gray Caulk	On Flashing - Sides (Main School Roof Above Library)	ND
20A,B	Dark Gray Caulk	Under Flashing (Main School Roof Above Library)	ND
21A,B	Gray Caulk	On Brick Wall On Flashing (Main School Roof Above Library)	ND
22A,B	Tar	On HVAC Units (Main School Roof)	ND
23A,B	Gray Caulk	On Bolts on HVAC Units (Main School Roof)	ND
24A,B	Gray Caulk	on HVAC Units (Main School Roof)	ND
25A,B	Gray Caulk w/ Black Coating	on HVAC Unit (Main School Roof)	ND
26A,B	Gray Caulk	Communication Tower (Main School Roof)	ND
27A,B	Dark Gray Caulk	on Mitsubishi Electric Unit Conduit (Main School Roof)	ND
28A,B	Black Window Glaze	School Entrance Roof	ND
29A,B	Dark Gray Window Caulk	School Entrance Roof	ND
30A,B	Black Caulk	On Flashing Between Window Units (School Entrance Roof)	ND
31A,B	White Caulk	Under Flashing	Chrysotile 5%

Sample ID	Material	Location	Asbestos%
		(School Entrance Roof)	
32A,B	Dark Gray Caulk	On Flashing/Bolts (School Entrance Roof)	ND
33A,B	Black Caulk	On Brick Conduit (School Entrance Roof)	ND
34A,B	Gray Caulk	On Flashing (Pool/Gyms) (Boiler Room)	ND

ND = Not Detected

Chrysotile = Asbestos Mineral

Refer to the table below for quantities of confirmed positive ACM and presumed ACM.

Material	Location	Quantity	Units
Gray Sealant on Gutter	Roof over Auditorium	100	LF
Tar (Vapor Barrier)	(Auditorium) On Wall	600*	SF
Tar	(Main School) Top of Deck	76,000**	SF
White Caulk	Under Flashing (School Entrance Roof)	20	LF

SF = Square Feet

LF = Linear Feet

EA = Each

* = The quantity includes entire curb associated with the auditorium roof due to homogenous material

** = The quantity includes the whole main school roof (excludes the main school entrance roof and two roofs connecting the library and main school) due to homogenous material. However, the tar does not cover the entire roof. The tar was not mopped onto the roof; it was spread throughout and used as an adhesive.

5.3 Recommendations

ACM that will be impacted by renovation or demolition work must be removed before they are disturbed. This work must be conducted in accordance with a project design as prepared by a licensed Asbestos Abatement Project Designer. This report is not intended for use as an abatement design. Prior to disturbance, the ACM identified must be abated by a Commonwealth of Massachusetts-licensed asbestos abatement contractor following all federal, state & local regulations governing asbestos abatement. A copy of the asbestos Waste Shipment record must be received within 35 days of removal from the Site. Asbestos air quality sampling must be conducted

under USEPA regulations following asbestos abatement and prior to re-occupancy of the spaces. If additional materials are discovered that have not yet been sampled, those materials should be considered ACMs until laboratory analysis determines otherwise. During renovation or demolition work, it is possible that additional suspect ACM will be encountered in areas such as subsurface tunnels, behind fixed walls, etc. and will need to be confirmed prior to disturbance. Contractors should conduct any such work in a controlled manner.

6.0 LEAD-BASED PAINT

6.1 Methods

CDW performed a visual inspection of painted surfaces. CDW collected samples from different color paints on various types of building component substrates. Samples were submitted to Aerobiology Laboratory Associates Incorporated, A Pace Laboratory, located in Woburn, Massachusetts for analysis via Atomic Absorption Spectrometry (AAS).

6.2 Findings

The results of the laboratory analysis are provided in the table below.

Sample ID	Sample Description/Location	Lead Concentration (% Weight)
LBP – 1	Maroon Paint on Louver – Roof over Swimming Pool	1.11
LBP – 2	Black Paint on Siding – Pool/Gyms Roof	0.132
LBP – 3	Black Paint on Siding – Auditorium Roof	<RL
LBP – 4	Black Paint on Siding – Main School Roof	0.113
LBP – 5	Blue Paint under Siding / On Columns – Main School Roof Entrance	<RL

<RL = Below the Reporting Limit

The Environmental Protection Agency (USEPA) and US Department of Housing and Urban Development definitions define LBP as any paint or surface coating that contains lead equal to exceeding one milligram per square centimeter (1.0 mg/cm²) or 0.5% by weight. The OSHA lead-in-construction standard defines lead containing paint (LCP) as a paint or coating containing any detectable level of lead. Five (5) total samples were collected for laboratory analysis. Three (3) out of the five samples collected reported detection of lead. Only one of the samples reported a concentration above 0.5% by weight, which makes sample LBP-1, Maroon Paint on Louver, considered LBP. The laboratory analytical report is included in Appendix C.

6.3 Recommendations

Based on the conclusions of this testing, the following recommendations are offered:

- During demolition or renovation, workers must be protected from exposure to any concentration of lead in paint materials. These surfaces should not be sanded, scraped, drilled, or otherwise altered unless proper engineering controls are used to prevent migration of fugitive lead containing dust from the work area. Under OSHA regulations, any demolition or renovation to be performed at a structure where lead in paint and dust is present must be performed in accordance with a worker protection policy, including, but not limited to, appropriate training, medical monitoring, respiratory protection, and other protective equipment.
- In addition to the worker protection requirements stipulated by OSHA, MassDEP and the USEPA regulate the disposal of wastes that are potentially hazardous. Such wastes may include paint chips and residue generated during abatement or repainting work, or whole components, such as wood windows, doors, and trim that are coated with LBP and that are disposed of as the result of renovation or demolition work. To determine the required method for disposing of permeable items coated with LBP, the MassDEP and the USEPA require representative sampling of the debris to determine the quantity of lead that would be expected to leach into the environment if the debris were disposed of in a landfill. The representative sample(s) must be analyzed by the Toxicity Characteristic Leaching Process (TCLP) to determine the proper disposal method.

Limitations

The conclusions are limited to the information available at the time of the field survey and the scope of services, as defined. No subsurface soil or groundwater sampling and analysis was performed. Where access to portions of the Site or to structures on the site was unavailable or limited, CDW renders no opinion as to the presence of hazardous material or the presence of indirect evidence related to hazardous material in that portion of the site or structure. This report cannot be solely relied upon for renovation or demolition. The sampling performed forms the basis for conclusions expressed and areas inaccessible for testing limits those conclusions. No other conclusions, interpretations or recommendations are contained or implied in this report other than those expressed. While CDW followed industry standards during the inspection, we do not warrant that all suspect hazardous building materials were identified in or on the buildings and shall not be held liable related to future abatement costs related to hazardous materials that are either not discovered or not appropriately characterized. This is due in part to inherent problems with every building inspection, such as, but not limited to:

- Seemingly homogeneous materials that are not in fact homogeneous;
- Seemingly representative locations that are not in fact representative;
- Layered materials that are not uniformly present or are isolated;
- Materials that are present and accessible but were not considered to be hazardous;
- Materials that are present in an isolated and limited quantity; and
- Material that is present in locations that are unsafe or otherwise difficult to access.



Client acknowledges that CDW's inspection is limited, and all hazardous materials may only become apparent during the course of future renovation or demolition. During the course of future renovation/demolition work, it is likely that additional hazardous materials or materials suspected of being hazardous will be identified. Such materials should be assumed to be hazardous unless appropriate evaluation or sampling and analysis demonstrate otherwise. No other use of this report is warranted without the written consent of CDW Consultants, Inc.

CDW appreciates the opportunity to provide our services to you on this project.

Very truly yours,
CDW Consultants, Inc.

A handwritten signature in black ink, appearing to read "Bryant Dana".

Bryant Dana, Asbestos Inspector
Environmental Scientist/Assistant Project Manager

A handwritten signature in black ink, appearing to read "Susan Cahalan".

Susan Cahalan, PG, ISSP-SA
Associate/Senior Environmental Geologist